



Over-fitting suppression training strategies for deep learning-based atrial fibrillation detection

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Received: 29 April 2020 / Accepted: 22 November 2020 / Published online: 2 January 2021
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Abstract

Nowadays, deep learning-based models have been widely developed for atrial fibrillation (AF) detection in electrocardiogram (ECG) signals. However, owing to the inevitable over-fitting problem, classification accuracy of the developed models severely differed when applying on the independent test datasets. This situation is more significant for AF detection from dynamic ECGs. In this study, we explored two potential training strategies to address the over-fitting problem in AF detection. The first one is to use the Fast Fourier transform (FFT) and Hanning-window-based filter to suppress the influence from individual difference. Another is to train the model on the wearable ECG data to improve the robustness of model. Wearable ECG data from 29 patients with arrhythmia were collected for at least 24 h. To verify the effectiveness of the training strategies, a Long Short-Term Memory (LSTM) and Convolution Neural Network (CNN)-based model was proposed and tested. We tested the model on the independent wearable ECG data set, as well as the MIT-BIH Atrial Fibrillation database and PhysioNet/Computing in Cardiology Challenge 2017 database. The model achieved 96.23%, 95.44%, and 95.28% accuracy rates on the three databases, respectively. Pertaining to the comparison of the accuracy rates on each training set, the accuracy of the model trained in conjunction with the proposed training strategies only reduced by 2%, while the accuracy of the model trained without the training strategies decreased by approximately 15%. Therefore, the proposed training strategies serve as effective mechanisms for devising a robust AF detector and significantly enhanced the detection accuracy rates of the resulting deep networks.

Keywords Atrial fibrillation (AF) · Electrocardiogram (ECG) · Deep learning · Wearable ECG

1 Introduction

Nowadays, smart wearable devices have become a hot topic, especially in the measurement of physiological signals [1]. Development of wearable electrocardiogram (ECG) devices makes the real-time and continuous individual ECG monitoring possible [2]. Atrial fibrillation (AF) is a very

common type of cardiac arrhythmia, characterized by three features: the absence of P waves, highly irregular variation of R-R intervals, and presence of undulating atrial activity, also known as fibrillatory waves or f waves [3–5]. AF may lead to stroke and congestive heart failure (CHF) and increase the death rate for AF patients [6]. Early diagnosis allows better treatment and prevention of secondary diseases like stroke [7]. Its reliable detection is an important target of long-term bio-signal monitoring and is still an unmet challenge even for clinical ambulatory ECG [4].

Traditional feature-based AF detection methods usually extract features from time-, frequency-, or amplitude-domains [8–10]. Features from the entropy perspective are also used for AF detection [11–16]. Deep learning-based models with unique hierarchical structures extract deep-level features and become new solutions to the analysis of biological signals [17–20]. There is also an increasing trend in the deep learning research for AF detection [21, 22]. Convolutional Neural Network (CNN) and Recurrent

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Neural Network (RNN) are two mainstream architectures of supervised deep learning methods.

CNN-based AF detection models can be simply divided into two categories: 1D CNN and 2D CNN models. The 1D CNN models were usually trained using ECG signal as input directly [23–25]. Researchers using 2D CNN-based models convert the 1D signal to a 2D representation. Xia et al. applied short-term Fourier transform (STFT) and stationary wavelet transform (SWT) to obtain the 2D matrix input suitable for deep 2D CNN models [26]. Qayyum et al. converted ECG signals into 2D images by STFT, and used pre-trained CNN models for transfer learning [27]. Lorenz plot imaging of ECG RR intervals was also used as input images to training a 2D CNN-based model for AF classification [4].

RNN is another type of deep networks, whose architectures allowing the network to retain and utilize state information of the input sequence. Thus, RNN is suitable for processing time series signals [28]. However, the traditional RNN may suffer from the issue of gradient vanishing or exploding, resulting a limited performance for long-term dependencies. Long short-term memory (LSTM) networks are a type of most widely used RNN architecture [29, 30]. Faust used LSTM-based deep learning model and RR interval signals to detect AF [31]. Chang et al. proposed an AF detection method which exploited the spectral and temporal characteristics of AF ECG signals with a multi-lead LSTM model [32]. Maknickas et al. proposed a three-layer LSTM deep learning model using pre-computed QRS complex features for classification [33]. Furthermore, deep learning models combining CNN with RNN were also proposed in past few years [34–38]. Some scholars have studied combined models using ECG signal as the input data [35–37]. Andersen et al. developed a CNN and LSTM combined deep learning classification model using RR interval as input [34].

All these deep learning methods had achieved promising AF detection performance on the commonly used MIT-BIH database, but whether these models can be applied to dynamic ECG data remains unknown since these data usually include more noise. Most of these deep learning methods utilized data from one database [4, 26, 39] and tested performance by the cross-validation method [27]. However, since the evaluation is conducted using a single data set, it is difficult to estimate how reliable a particular cross validation performance is [40]. In other words, it is hard to evaluate the over-fitting degree of the developed models. Therefore, how to reduce the over-fitting and improve the generalization ability of deep learning model is a very important issue [21]. Chang et al. performed AF detection and obtained an accuracy of 85% on their test set, while their model achieved an accuracy of 98.3% in the training phase, indicating the huge difference between

training and testing, i.e., over-fitting [32]. The network proposed by Andersen achieved an accuracy of 97.80% on the training set for AF detection, as well as a significantly decreased accuracy of 87.40% on new recordings [34]. The over-fitting of these models is mainly caused by two factors:

- a. the intra-subject similarity between different ECG signals collected from one person, which can lead to the trained model extracted more personal features beyond AF [34].
- b. the inter-subject difference between different patients, which results in the sharp drops in accuracy rates when the model tested on an unseen database.

Therefore, in this study, we proposed two strategies to improve the robustness and generalization capabilities of the model. The first training strategy utilizes an Fast Fourier transform (FFT) and Hanning-window-based filter to remove the individual difference in the ECG waveforms. By so doing, the ECG data collected from different patients hold a similar property of waveform characteristics after pre-processing. Therefore, such a process is able to attenuate the individual waveform difference between patients. The second training strategy employs wearable ECGs as the training data to improve the robustness of the proposed model to the dynamic ECG data. By import ECG data of wearable ECGs, the trained model can extract robust features and reduce the noise interference. And the so trained model could better adapt to noisy ECG signals in dynamic applications. Twenty-nine patients have been monitored by wearable devices for at least 24 h with no activity restrictions. To verify the effectiveness of the training strategies, an LSTM and CNN-based detection model was proposed and test on separate databases. The ECG signals from 20 patients are extracted for model training, while the ECG signals form the rest of the patients are used as the separate test dataset. Additional datasets such as the MIT-BIH Atrial Fibrillation Database (AFDB) and the PhysioNet/Computing in Cardiology Challenge 2017 dataset (CinC 2017) are also employed for model evaluation.

2 Datasets

2.1 Wearable ECG databases I and II

The wearable ECG data were collected by a wearable ECG monitoring device developed by the authors' team and Lenovo [2], with a sampling frequency of 400 Hz and 16-bit resolution over a range of ± 10 mV. The ECG signals collected of standard lead I were stored in the ECG module and subsequently uploaded to the cloud server via WiFi module. Twenty-nine patients aged 26 to 65 participated

in data collection, 11 patients suffered from AF, and four had a history of premature contractions. A set of 24-h ECG signal was collected from each patient without any activity restrictions. Then, the collected ECG episodes were labeled as AF and non-AF classes by Cardiologists. Three cardiologists participated in the annotation process, two of them annotated independently. And the doubtful cases were rechecked by the third cardiologist for final decisions.

ECG waveforms from 20 patients (10 with AF, 2 with premature contractions, and 8 without abnormal heart rhythm) were selected as wearable dataset I (wearable I). The remaining ECG signals from 9 patients (1 with AF, 2 with premature contractions, and 6 without abnormal heart rhythms) were selected as the wearable dataset II (wearable II). In this study, the ECG signal with premature contractions was labeled as non-AF class. The gender and age range information of these patients was showed in the following Table 1.

The ECGs in the wearable I were segmented into 10 seconds segments and divided into 10-fold without overlapping for the evaluation using cross-validation. Because of the similarity between ECG segments of one same person, only 4000 ECG segments of each patient were randomly chosen. And in each fold of the tenfold cross-validation, segments of 9 patients with AF and other 9 patients without AF were used as training data, and segments of the remain 2 patients (1 with AF and 1 without AF) were set as validation data. Thus, a total of 80,000 ECG segments were extracted. For the 10-fold cross-validation, 72,000 segments were used as the training set and the remaining 8000 were used as the validation database for training phase. All ECG segments from the 9 patients in the wearable II were extracted and used as the test set.

2.2 The MIT-BIH atrial fibrillation database

The AFDB consists of 25 long-term ECG recordings of human subjects with AF (mostly paroxysmal) [41]. Each recording is 10-h duration and contains two leads of ECG signals sampled at 250 Hz. The rhythm annotation files were prepared manually; they contain rhythm annotations of

the following types, i.e., “AFIB” (atrial fibrillation), “AFL” (atrial flutter), “J” (AV junctional rhythm), and “N” (all other rhythms).

In this study, ECG signals labeled as “J,” “N,” and “AFL” were referring to the non-AF ECG data, while the signals labeled as “AFIB” were used as the AF ECG samples. Moreover, we only utilized the second lead of the ECG signal in this database and all the ECG signals were resampled to 400 Hz. In some ECG segments of MIT-AFDB, there were large differences in the QRS waveform between the different two leads, which also affects the waveform of the corresponding position of the signal after preprocessing. And the proposed model was designed to overcome the waveform changes and catch the arrhythmia features. Thus, ECGs from the second lead of MIT-AFDB would be a good test data to evaluate the overfit in the trained models which was trained only by the lead I ECG data. Then the ECG signals with the length over 10 s (4,000 points) were divided into several 10-s segments with no overlapping. Meanwhile, the last 10-s signal of each ECG sub-waveforms was also added to make sure no data missing after division. After segmentation, we obtained 33,484 AF and 49,980 non-AF segments, respectively.

2.3 The physionet/computing in cardiology challenge 2017 database

The ECG recordings in the CinC 2017 database are collected using the AliveCor device, where the training set contains 8528 single lead ECG recordings lasting from 9 s to slightly over 60 s. All the ECG segments are collected from standard lead I. The ECG recordings pertaining to the Challenge 2017 dataset were sampled as 300 Hz and have been band pass filtered by the AliveCor device [3]. The dataset contains 4 different types of rhythms, i.e., AF (atrial fibrillation), normal (normal rhythm), other (all other rhythms), and noise.

In this study, ECG data labeled as noise in the CinC 2017 dataset were abandoned, while normal and other rhythm data were used as the non-AF class with AF rhythm data used as the AF class. All the recordings were resampled

Table 1 The age ranges and gender of patients in wearable databases I and II

Age	Wearable database I				Wearable database II			
	AF		Non-AF		AF		Non-AF	
	Male	Female	Male	Female	Male	Female	Male	Female
26–40	–	–	1	1	–	–	2	3
40–55	2	2	2	3	1	–	–	1
55–65	4	2	2	1	–	–	1	1
Total	6	4	5	5	1	–	3	5

to 400 Hz. By the way, the ECG signal parts longer than 10 s (4000 points) were also divided into several 10-s segments without overlapping and the last 10-s data of each recording was added. In addition, the ECG segments were re-annotated by Cardiologists. Finally, a total of 1821 AF and 17,399 non-AF segments were obtained.

3 Method

3.1 Pre-processing operations

Since the wearable ECG signals usually contain different kinds of noise, band pass filter was a good method to alleviate the impact of noise. In this research, a Butterworth-bandpass filter with a passband of 0.5–45 Hz was used to remove the baseline drift and reduce the high-frequency noise.

The inter-subject difference in ECG signals between different individuals is usually manifested in the following two aspects: The morphology difference of the QRS complex and the overall amplitude range difference of the ECG signals. The FFT and Hanning-window-based filter was used to reduce the individual differences of QRS complex. ECG segments were transformed by FFT firstly. Then, the frequency range of 1–45 Hz was intercepted by Hanning window for each FFT-transformed signal, and the Inverse Fast Fourier Transform (IFFT) was performed to obtain the filtered signal.

As shown in Fig. 1, the sub-figures in the left column (A, B, and C) were three ECG segments with obvious rhythm and waveform differences; the sub-figures in the right column (D, E, and F) were the corresponding ECG segments after eliminating the individual differences in the QRS complex.

The minimum-maximum normalization method was adopted to amplify the filtered ECG segments into

the interval (0, 1). Thereby, the FFT and Hanning-window-based filter and normalization method removed the morphology difference of QRS complex and the overall amplitude difference in ECG signals between patients.

3.2 Network structure and optimization method

As showed in Fig. 2, the classification model proposed in this study was mainly composed of three components: LSTM, 1D-CNN, and Softmax classification modules. The ECG data after pre-processing were firstly passed through a 1D maxpooling layer with a stride size of 8 to reduce the signal processing time length and reduce the interference of small amplitude noise on the signal. By so doing, the input of the LSTM module was transformed to a time series with a length of 500. The output dimension was set to 200, and the output at each time step was recorded as a feature extracted from the input ECG segment.

The 1D CNN module consists of three 1D CNN Blocks, where each CNN block consists of four layers: a 1D convolutional layer, a batch normalization layer, a 1D-maxpooling layer, and an activation layer with activation function of tanh. Each block uses 32 convolution kernels, and the convolution kernel size is set to 3 for each convolution layer. The last CNN block did not use 1D-maxpooling layer. The features extracted by 1D CNN were stretched into a feature sequence of 4000 points by the flatten layer. Then the feature sequence was classified into AF or non-AF classes by the softmax layer.

The proposed LSTM-CNN model used a cross-entropy loss function and was trained by the Adam optimization method. In order to prevent over-fitting, the batch normalization layer was added after the LSTM layer and each convolution layer, and the dropout mechanism was added after each convolution layer and the fully connected layer. Meanwhile, the L2 regularization method of the fully connected layer was also implied to improve the model generalization

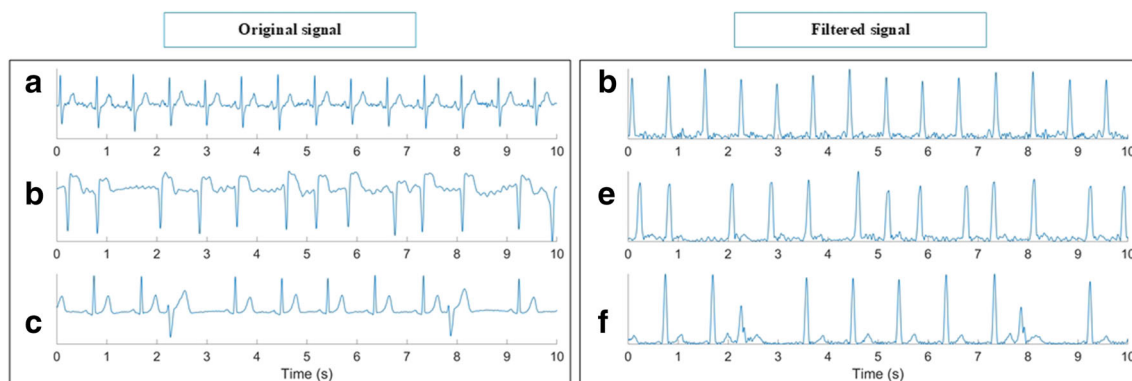


Fig. 1 ECG signals from different subjects and pre-processed ECG signals. (A), (B), and (C) ECG segments from different patients, where (A) was the ECG signal of normal rhythm; (B) was the ECG signal of

patient with atrial fibrillation; and (C) was the ECG signal of patient with premature beats. (D), (E), and (F) The corresponding signals after pre-processing

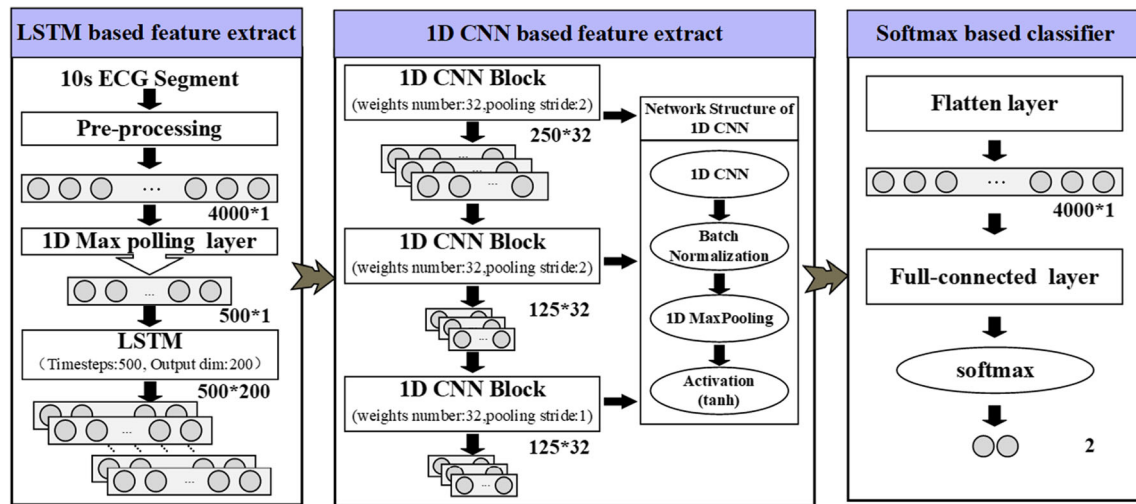


Fig. 2 The proposed classification model based on LSTM and CNN

ability. The early stopping training method was also adopted to reduce over-fitting.

3.3 Model performance evaluation mechanism

In order to measure the model performance, the sensitivity (S_e), specificity (S_p), and accuracy (A_{cc}) were calculated as:

$$S_e = \frac{TP}{TP + FN} \times 100\% \quad (1)$$

$$S_p = \frac{TN}{TN + FP} \times 100\% \quad (2)$$

$$A_{cc} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (3)$$

where TP represents the number of AF segments which were correctly predicted as AF class; FN represents the number of AF segments which were falsely predicted as non-AF class; FP represents the number of non-AF segments which were falsely predicted as AF class; TN represents the number of non-AF segments which were correctly predicted as non-AF class.

4 Experimental result

4.1 Evaluation results of the proposed model adopting FFT and Hanning-window-based filter

To evaluate the effectiveness of the proposed training strategies, three models with the same network structure were yielded. As shown in Fig. 3, in order to verify the effectiveness of the FFT and Hanning-window-based filter, model B was trained with filtered ECG data while model A trained with ECG data which were not filtered. Then the

10-fold validation was performed and the test results were illustrated in Table 2.

The average validation accuracy, sensitivity, and specificity scores of model A were 98.91%, 98.50%, and 99.42%, respectively, while the average validation accuracy, sensitivity, and specificity scores of model B were 98.92%, 98.87%, and 99.12%, respectively. On the contrary, when tested on wearable datasets, accuracy significantly decreased. The test accuracy of model B was 89.99% on the CinC 2017 challenge database and 88.10% on the wearable database II. Model A only achieved an accuracy of 84.67% on the CinC 2017 challenge database and an accuracy of 83.38% on the wearable database II. With respect to the comparison of the cross-validation results on the training set, the accuracy rates were reduced by about 10% for model B and 15% for model A.

4.2 Evaluation results of the proposed model using wearable ECG as the training data

In order to verify the effectiveness of using wearable ECG as training data, model B was trained on AFDB while model C was trained on wearable database I. Both two models adopted the FFT and Hanning-window based filter in pre-processing procedures. In the test phase, model B was tested on CinC 2017 challenge database and wearable database II. Model C was tested on wearable database II, AFDB, and CinC 2017 challenge database.

Model C achieved an average validation accuracy of 97.71%, sensitivity of 98.13%, and specificity of 97.29% in the training database, as well as obtained an excellent performance on the test databases. The test accuracy of model C was 96.23% on the CinC 2017 challenge database and 95.44% on the wearable database II. Model C was also tested on AFDB with an accuracy of 95.28%. There was

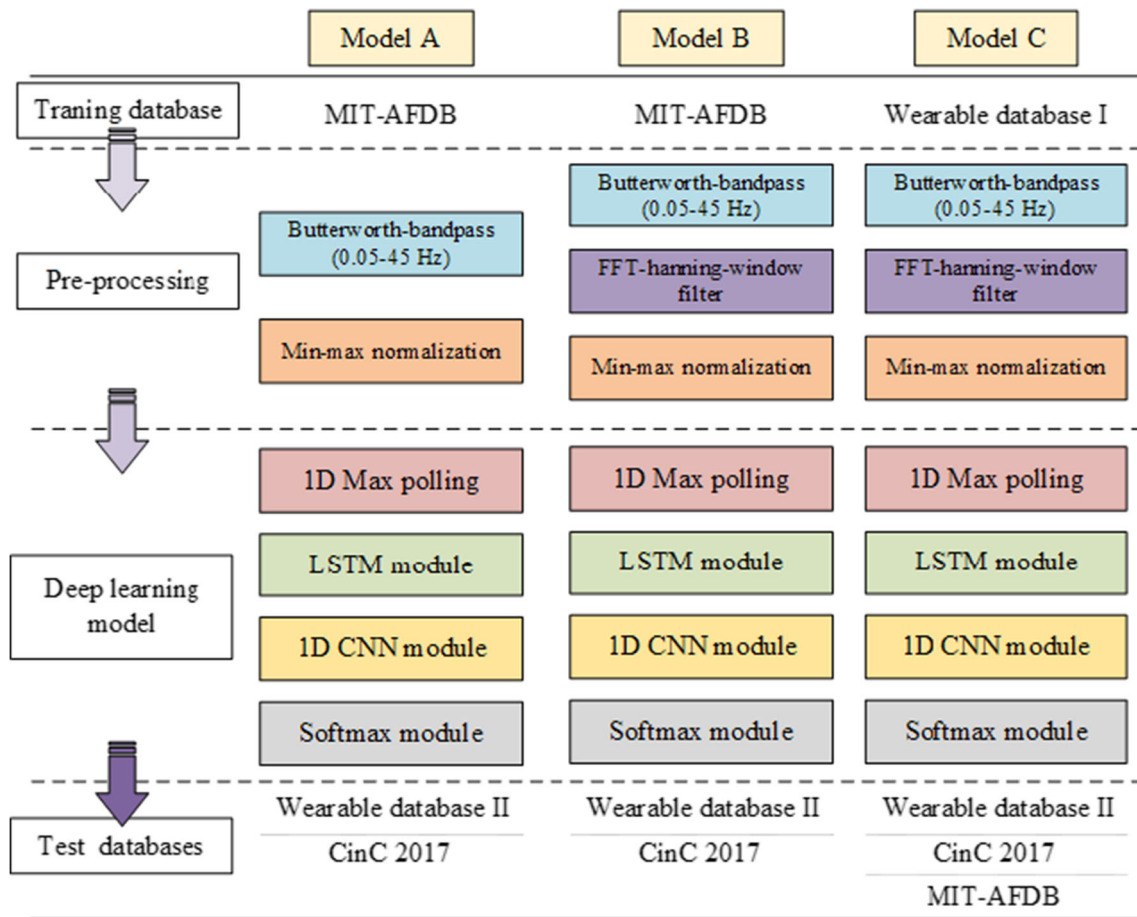


Fig. 3 The training and test flowchart of the three different instances of the proposed model

only about 2% decrease in accuracy rates when model C tested on separate databases, in comparison with the results obtained in the training stage. The test results indicate that although model C obtained the worst validation accuracy scores, model C maintained a good generalization ability for the test sets.

5 Discussion

5.1 Effectiveness of the proposed training strategies

In this study, we proposed two training strategies to improve the robustness and the generalization capability of the

Table 2 Performance of three different models on validation and test databases

Model	Method	Training set	Validation performance			Test set	Test performance		
			$S_e(\%)$	$S_p(\%)$	$A_{cc}(\%)$		$S_e(\%)$	$S_p(\%)$	$A_{cc}(\%)$
Model A	LSTM+CNN	AFDB	98.50	99.42	98.91	CinC 2017	75.67	85.61	84.67
						Wearable II	92.64	82.37	83.38
Model B	Filter LSTM+CNN	AFDB	98.87	99.12	98.92	CinC 2017	79.79	91.06	89.99
						Wearable II	91.40	87.75	88.10
Model C	Filter LSTM+CNN	Wearable I	98.13	97.29	97.71	CinC 2017	92.09	96.66	96.23
						Wearable II	97.73	95.19	95.44
						AFDB	96.46	94.49	95.28

trained model. One strategy was that a FFT and Hanning-window-based filter was utilized to remove the individual difference in the ECG waveform. Models A and B were trained with same ECG segments, but the ECG data of model B were filtered in pre-processing. Models A and B obtained approximately the same cross-validation results on the training database with the average validation accuracy of 98.91% and 98.92%, respectively. However, there was a nearly 5% accuracy improvement when adopting the proposed filter in pre-processing. Therefore, the proposed filter plays an important role in overcoming over-fitting.

The other training strategy was that using wearable ECG segments as training data to improve the robustness of the proposed architecture. Models B and C were trained with the same pre-processing method but different ECG segments from different databases. Model C was trained with wearable ECG data while model B was trained by ECG recordings from AFDB. The test accuracy of model C on two wearable datasets was 96.23% and 95.44%, which were close to the validation accuracy. However, the test accuracy of model B on wearable databases II and CinC 2017 database drew down around 10% compared to its validation accuracy of model B. Therefore, it can be concluded that the model trained with wearable ECG data was more robust than the network trained with ECG signals from traditional databases.

5.2 Classification capability of different deep learning models

In this work, the proposed LSTM-CNN-based model shows a relatively similar result on AFDB compared to other models proposed by Zhou [24], Xia [26], Wang [23], Andersen [34], and Faust [31]. As an example, the accuracy of the proposed model with input of 4-s ECG was 97.4% [23] while the model proposed by Xia achieved the accuracy score of 98.29% with input of 5-s ECG [26]. The accuracy

of our model with input of 10-s ECG was 98.92%, and the model proposed by Chang[32] achieved 98.50%. The best accuracy result of 99% was achieved by the model proposed by Zhou [24] with input of 30-s ECG signals. It demonstrates a trend that the accuracy of the model becomes higher when longer ECG segments were used.

RR interval-based method proposed by Andersen [34] showed 98.96% specificity and 86.04% sensitivity on new recordings. RR interval-based method proposed by Faust [31] also showed excellent performance on unseen data with an accuracy of 99.77%. However, these RR interval-based models require longer ECG segments, which implies that these models may not be suitable for undertaking real-time analysis. Compared to these methods of converting ECG segments to spectrograms as 2D matrix, our method used 1D ECG segments as input and directly reduced the input size and computational complexity. Compared to other 1D CNN models proposed by Wang [23] and Zhou [24], our model used 1D max-pooling method to reduce the input length. The input size of our model was only 500×1 which contains information of 10-s ECG, and achieved a similar result to that of the model proposed by Zhou [24]. Therefore, our model achieved the second best result with the shortest input length. Although longer ECG input will improve the classification accuracy, it will greatly increase the consumption of computing resources and lead to sharply drop of the predict speed. Using 10-s ECGs as model input can reduce computing time and improve real-time performance under the premise of satisfactory accuracy. Thus, 10 s would be an ideal input length for AF detection models.

5.3 Generalization ability of different deep learning models

As shown in Table 3, some previously proposed deep learning models were also tested on separate databases or

Table 3 Performance of methods tested on independent data or separate database

Author	Method	Training set	Validation performance			Test set	Test performance		
			$S_e(\%)$	$S_p(\%)$	$A_{cc}(\%)$		$S_e(\%)$	$S_p(\%)$	$A_{cc}(\%)$
Limam[42]	CNN+LSTM	CinC 2017 (85%)	82.5	98.70	90.60	CinC 2017 (15%)	72.70	98.60	85.60
Aderson[34]	30 RR+ LSTM+CNN	AFDB	98.98	96.95	97.80	MITDB NSRDB	98.96	86.04	87.40
Chang[32]	STFT+LSTM	Six databases	97.80	99.20	98.50	Separate data	86.88	79.55	83.21
Proposed	Filter+ LSTM+CNN	Wearable I	98.13	97.29	97.71	CinC 2017	70.17	–	75.60
						Wearable II	92.09	96.66	96.23
						AFDB	97.73	95.19	95.44
							96.46	94.49	95.28

independent ECG data. Limam[42] used 15% ECG signals from CinC 2017 database as independent test data, and the accuracy of the model on the test set was 85.60% which was about 5% lower than the result on the training set. Aderson[34] tested their model on two independent databases, and the accuracy dropped by about 10%. The model proposed by Chang[32] obtained an accuracy of 83.21% on separate data from six databases and 75.60% on CinC 2017 database while the model achieved an validation accuracy of 98.50% in training phase. In comparison with these existing studies, the test results of our model trained with the proposed training strategies only reduced by 1% or 2% on separate databases. Thus, the test performances ascertain the efficiency of the proposed training strategies, which effectively improve the robustness and generalization ability of the proposed model.

6 Conclusion

In this research, we proposed two training strategies to overcome the over-fitting in deep learning models. The FFT and Hanning-window-based filter was used to remove the individual differences in the ECG waveform. Wearable ECG signals which contained complex noise were also used in the training phase to improve the robustness of the proposed model. The proposed LSTM-CNN model was tested on three different ECG databases and the results on all test databases indicate that the proposed model maintained good generalization capability when tested on separate databases. The comparative experiments also illustrate that the above-mentioned training strategies can effectively improve the robustness and generalization abilities of deep networks for AF detection. Therefore, these selected training strategies can serve as effective alternative methods of eliminating over-fitting for future deep learning-based AF detection models.

Funding This study was supported by the National Key Research and Development Program of China (2019YFE0113800), the Distinguished Young Scholars of Jiangsu Province (BK20190014), the National Natural Science Foundation of China (81871444, 62001105 and 62071241), the Primary Research & Development Plan of Jiangsu Province (BE2017735), and the Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX19.0068).

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Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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